

Section: ME 423 X2

Individual Concept and Subsystem Analysis:

User-Side Load-Induced Deformation and Potential F/T Measurement Error

Team: Robotic Key

Name: Max Yao

"I pledge my honor that I have abided by the Stevens Honor System."

Subsystem and Concept Selection

Subsystem Analyzed:

This report evaluates the printed PLA handle and handle-to-sensor adapter on the user side of the selected instrumented handheld end-effector. The design question is whether operator-applied forces and moments can deform this user-side structure or its mounting region enough to threaten the stiff mechanical boundary needed for trustworthy force/torque (F/T) data.

Phase 1 Concept Selected:

The Phase 1 concept screening selected DF-1, the instrumented handheld end effector, because it balanced portability, natural manipulation, and direct F/T sensing. The Phase 1 report also selected HH-1, a one-hand motorized end effector with button control, over alternatives including the passive/fixed-key HH-4. HH-4 was simpler, lighter, and lower cost, but was set aside because it was less useful for future manipulation tasks and modularity [1].

Subsystem Selection Justification

The user-side structure directly supports Phase 1 needs N9 (minimize F/T contamination), N15 (minimize operator fatigue), N16 (F/T sensor as the primary stiff structural load path), N26 (manufacturable with available additive/COTS methods), N29 (support F/T calibration/validation), and N31 (safe repeated human operation) [1]. Phase 1 also identified the thickness/infill of printed PLA components as an area requiring Phase II technical analysis [1].

Phase 1 Traceability, Metrics, and Targets

Table 1 distinguishes targets already stated in Phase 1 from preliminary detailed-design criteria introduced here to convert qualitative Phase 1 needs into testable subsystem checks. The new criteria are therefore not substitutes for Phase 1 requirements; they are engineering acceptance criteria used to evaluate this subsystem.

Metric	Phase 1 trace	Units	Target / acceptance criterion	Basis	Current status
Structural stress under user-side load	N16, N18, N31	[MPa]	$\sigma_{vM} \leq 12.5$ [MPa]	Derived: provisional PLA S_y=25 [MPa] with screening n=2	10.32 [MPa] max → PASS
User-side structural displacement	N8, N9, N16, N29	[mm]	$\delta_{max} \leq 1.0$ [mm]	Preliminary detailed-design stiffness target, minimize contamination / permit precise motion	0.546 [mm] max → PASS
F/T contamination from upstream-only disturbance	N9, N29	[N]; [N·m]	Ideal $\Delta F=0$, $\Delta\tau=0$; minimize in practice	Phase 1 need; requires physical sensor test	NOT VALIDATED by FEA
Handled system mass / fatigue-safety	N15, N17, N31	[kg] ([lb])	System <1.361 [kg] (3.00 [lb])	Explicit Phase 1 target	Handle=0.05589 [kg] =4.1 [%] of budget
Additive manufacturability	N26	[%] of custom user-side structural parts	100 [%] printable with available process	Operationalization of Phase 1 manufacturability need	2/2 parts =100 [%] → PASS

Table 1. Design metrics traced to the Phase 1 stakeholder needs and target specifications. Structural stress and displacement thresholds are detailed-design screening criteria; the 3 [lb] system-mass limit is an explicit Phase 1 specification [1].

Detailed Design and Modeling Assumptions

Geometry and Load Path:

The analyzed structure contains the current printed handle and printed adapter. The adapter mounts to the user-side flange of the ATI Gamma through an eight-hole pattern. The Gamma itself is omitted from the simplified structural model; its support is represented with two fixture idealizations. The handle-to-adapter interface is treated as bonded so the study isolates printed-part compliance and the sensor-side support condition.

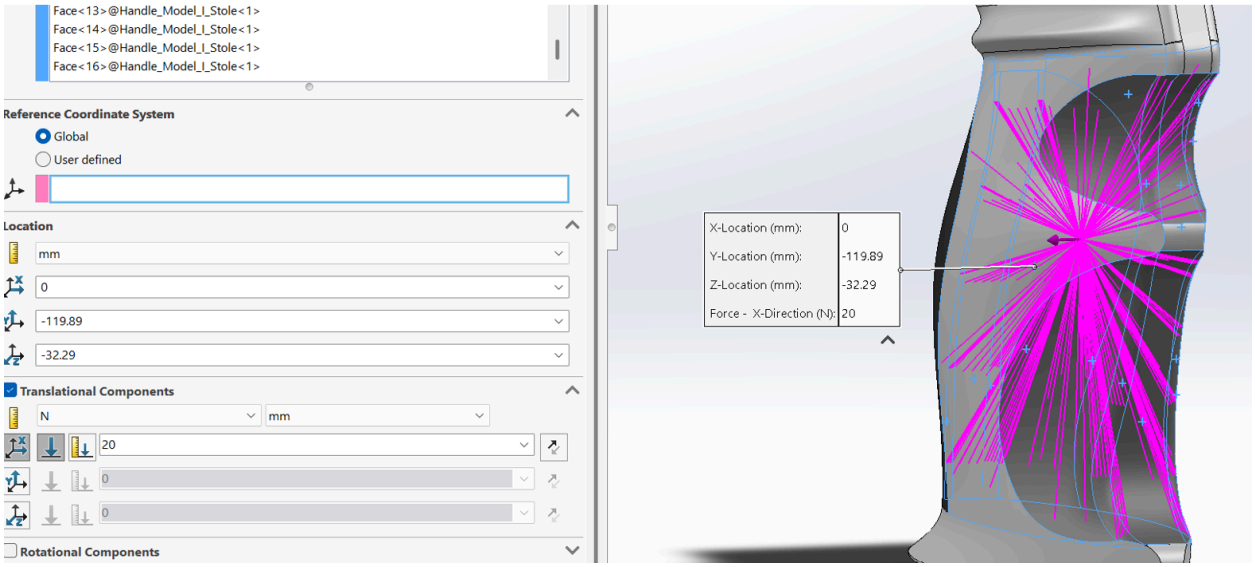


Figure 1. SolidWorks distributed remote-load setup. The resultant load is applied near the center of the gripped region rather than to a single finger-groove face.

Measured Density

The handle mass is 55.89 [g] and CAD volume is approximately 146,678.302 [mm³]. Because the adapter uses the same print settings, the measured handle is used to define an effective printed density for both parts:

$$\rho_{effective} = \frac{m}{V} = \frac{0.05589\ [kg]}{1.46678 \cdot 10^{-4}[m^3]} \approx 381.0379\ [\frac{kg}{m^3}]$$

Assumption / input	Value or treatment	Reason / limitation
Material model	Linear elastic, isotropic effective FDM PLA	Actual FDM PLA is anisotropic; screening approximation only
Elastic modulus, E	1500 [MPa]	Conservative preliminary effective stiffness
Poisson ratio, ν	0.35	Isotropic approximation
Shear modulus, G	556 [MPa]	Computed from E/[2(1+ν)]
Yield strength, S_y	25 [MPa]	Provisional screening value; not coupon-validated
Density	381 [kg/m³]	Derived from measured handle mass/CAD volume
Handle-adapter interface	Bonded	Joint-specific compliance not evaluated
Sensor-side fixture A	Entire adapter face fixed	Stiff upper-bound support
Sensor-side fixture B	Eight cylindrical screw-hole regions fixed	Discrete-support comparison; still idealized
Gravity	Included	Uses effective density
Mesh quality	Preliminary mesh; no formal convergence study	Peak local stress may remain mesh-sensitive

Tool-side loads	Excluded	Study isolates user-side loading only

Table 2. Modeling assumptions and limitations used in the static studies.

Engineering Analysis

Simplified Theoretical Benchmark

Before interpreting the finite-element model, the user-side structure was reduced to an intentionally simple cantilever benchmark. This model is not fitted to the detailed CAD; it is used only to check whether the selected 20 [N] and 1.0 [N·m] reference loads produce plausible stress and deformation magnitudes. A 120 [mm] long, 20 [mm] equivalent circular section is used with E = 1500 [MPa] and G = 556 [MPa].

For d = 20 [mm], the section properties are:

$$I = \frac{\pi d^4}{64} = 7854\ [mm^4]$$

$$J = \frac{\pi d^4}{32} = 15708\ [mm^4]$$

Shigley gives the standard cantilever relationships:

$$M = FL\ ,\ \sigma = \frac{Mc}{I}\ ,\ \delta = \frac{FL^3}{3EI}$$

While torsion is screened with:

$$\tau = \frac{Tc}{J}\ ,\ \phi = \frac{TL}{GJ}$$

Accordingly, setting a 20 [N] transverse load results in:

$$M = 20\ [N] \cdot 120\ [mm] = 2.40\ [N \cdot m]$$

$$\sigma_b = \frac{Mc}{I} = 3.06\ [MPa]$$

$$\delta = 0.978\ [mm]$$

$$\tau_{max} = 0.637\ [MPa]$$

$$\phi = 0.788\ [\text{deg}]$$

$$\sigma_{VM} = \sqrt{\sigma_b^2 + 3\tau^2} = 3.5\ [MPa]$$

The simplified 20 [N] bending case remains below the 12.5 [MPa] screening allowable and produces approximately 1 [mm] deformation, so the selected reference load is large enough to exercise the design without being an obviously destructive input in the idealized model.

SolidWorks Static-Study Method

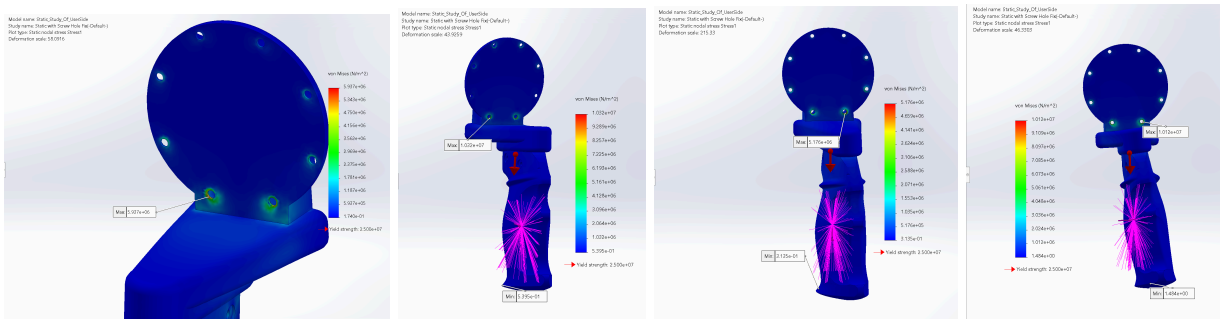
Four quasi-static user-side load cases were applied through the distributed remote point near the grip center: (A) 20 [N] in global X, (B) 20 [N] in global Z, (C) 1.0 [N·m] about the approximately grip-aligned Y axis, and (D) 20 [N] in X combined with 1.0 [N·m] about Y. Gravity was included. Each case was solved with both fixture models from Table 2.

The full-face fixture intentionally represents a very stiff mounting limit. The eight-hole fixture allows the plate between fastener locations to deform and therefore better exposes stress localization near the screw pattern, but it still does not represent finite screw stiffness, preload, friction, thread engagement, or interface separation. Shigley specifically cautions that FEA boundary conditions can dominate results and recommends checking alternate limiting assumptions when the real support is uncertain [2].

Results and Evaluation

Load case	Full-face stress [MPa]	Full-face URES [mm]	8-hole stress [MPa]	8-hole URES [mm]	8-hole screening n=25/σ
20 [N] X	2.012	0.2808	5.937	0.3766	4.21
20 [N] Z	1.581	0.4052	10.32	0.5462	2.42
1.0 [N·m] Y	0.944	0.0612	5.176	0.1025	4.83
20 [N] X + 1.0 [N·m] Y	1.856	0.2794	10.12	0.4725	2.47

Table 3. Completed SolidWorks results. URES is resultant displacement. The last column is a screening ratio based on the provisional 25 [MPa] yield value; it is not a validated material factor of safety, but is a simpler check.



Figures 2-5: Eight-hole von Mises stress contours for the four completed load cases. Peak stresses localize near the lower mounting holes in the more severe cases.

Governing Case

The 20 [N] Z-direction load with the eight-hole fixture governs the completed study at 10.32 [MPa] and 0.5462 [mm]. The combined 20 [N] X + 1.0 [N·m] Y case is nearly as severe at 10.12 [MPa] and 0.4725 [mm]. Both satisfy the detailed-design structural targets $\sigma \leq 12.5$ [MPa] and $\delta \leq 1.0$ [mm].

Boundary-Condition Sensitivity

Changing from a fully fixed face to fixed screw-hole regions increases peak stress by approximately 195 [%] (X), 553 [%] (Z), 448 [%] (Y torque), and 445 [%] (combined). Maximum displacement increases by approximately 34 [%], 35 [%], 67 [%], and 69 [%], respectively. The mounting assumption therefore has a first-order effect on the result; the full-face model is too stiff for final verification, while the eight-hole model is useful as a conservative screen but may exaggerate local stress at the idealized fixed hole walls.

Comparison with Theory

The simplified cantilever predicts 3.06 [MPa] and 0.978 [mm] for a 20 [N] transverse load. These values are the same order of magnitude as the FEA, but the detailed CAD and discrete support produce different local stresses and directional stiffness. The theoretical model therefore supports the reasonableness of the selected load magnitude without being used to fit or validate local FEA peaks.

Broader Impacts

Public Health/Safety and Operator Fatigue

Phase 1 specifies a total system-weight target below 3.00 [lb] (1.361 [kg]) and identifies minimized fatigue and safe repeated human operation as needs N15 and N31 [1]. The measured handle mass is 55.89 [g] (0.123 [lb]), which consumes only 4.1 [%] of the total mass budget and leaves 1.305

[kg] (2.88 [lb]) for the adapter, sensor, end effector, camera, fasteners, wiring, and other hardware. This does not validate the final system mass, but it shows that the handle itself is not a dominant contributor to the Phase 1 weight/fatigue constraint.

Manufacturing

Need N26 requires the system to be manufacturable using available machining, additive manufacturing, and COTS components [1]. Both custom user-side structural parts evaluated here – the handle and adapter – are produced by the available FDM process, giving 2/2 = 100 [%] additive manufacturability for this custom subsystem. The tradeoff is that material anisotropy and print-process variability become design uncertainties that must be addressed through coupon testing or conservative screening properties.

Verdict, Limitations, and Next Validation

The governing completed case produces 10.32 [MPa] maximum stress, below the 12.5 [MPa] screening allowable, and 0.5462 [mm] maximum displacement, below the 1.0 [mm] preliminary stiffness target. The handle also represents only 4.1 [%] of the Phase 1 system-mass budget, and both custom user-side structural components are manufacturable with the available additive process.

However, Phase 1 requirement N9 is not yet validated. These simulations do not calculate the Gamma sensor's electrical F/T error or prove that an upstream hand disturbance produces zero parasitic wrench. They only show that the present printed structure remains below the selected stress/displacement screening limits for the tested reference loads. The large difference between fixture models confirms that the real eight-screw joint is the primary mechanical uncertainty.

Future verification should (1) replace the ideal fixture with an eight-screw contact/preload model once the fastener and tightening data are finalized, (2) perform a mesh-convergence study around the holes and fillets, and (3) physically hold downstream pose/contact constant while applying controlled user-side forces and torques. The resulting change in the measured wrench, is the experiment that can directly validate N9.

AI Use and Independent Verification

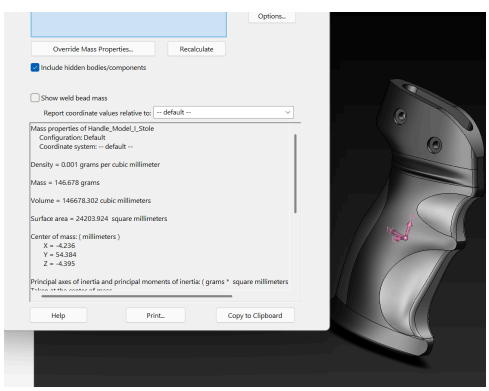
A LLM was used to keep track of the assumptions and help with report organization. The SolidWorks study geometry, load cases, fixture definitions, and result screenshots were generated independently. Governing beam, torsion, and FEA-validation relationships were cross-referenced to Shigley [2], and the numerical SolidWorks outputs were compared against independent closed-form order-of-magnitude calculations.

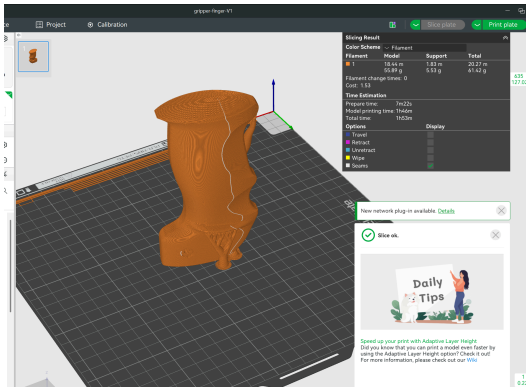
References

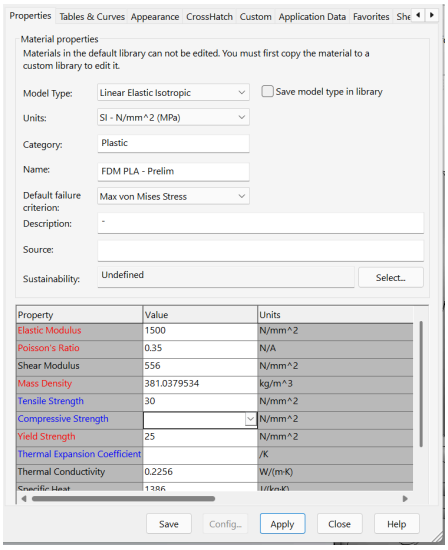
[1] Team C4, "Design of an Instrumented Robot Data-Collection Cell for Physical AI," ME 423 X2 Phase I Report, 2026.
[2] R. G. Budynas and J. K. Nisbett, Shigley's Mechanical Engineering Design, 11th ed. New York, NY: McGraw-Hill Education, 2020, especially Chs. 4, 8, and 19

Appendix

Group A.1 - Acquiring Printed Density & Custom PLA Material Stand-In:

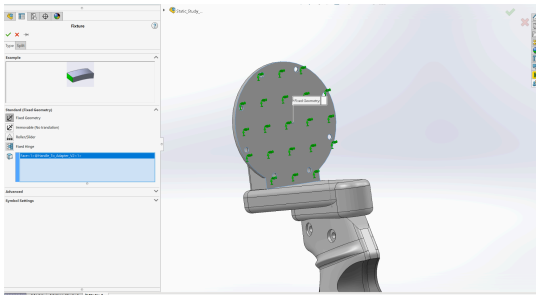




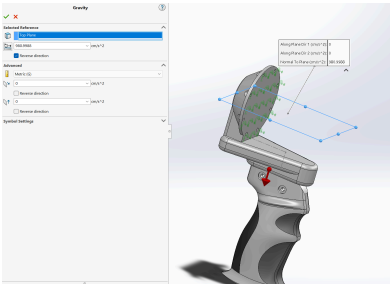


Group A.2 – Setup and Results of SolidWorks Simulations:

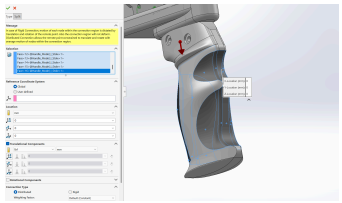
Face Fixture:



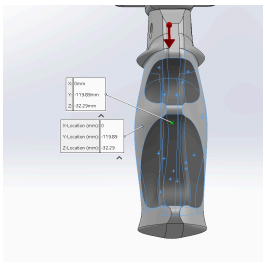
Handle Load as a Remote + Distributed Load:



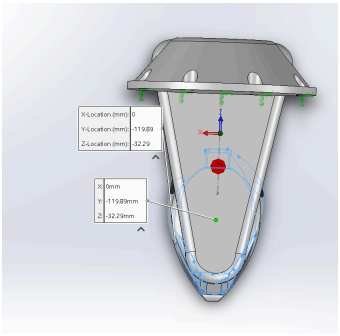
Handle Load as a Remote + Distributed Load:



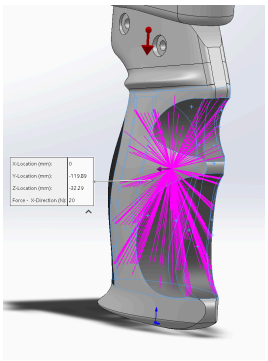
Approximating the Center of Operator Load (1):



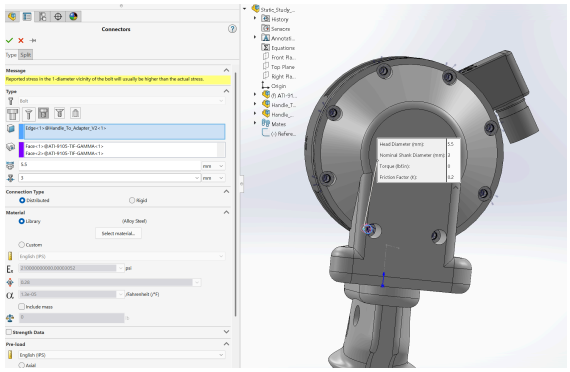
Approximating the Center of Operator Load (2):



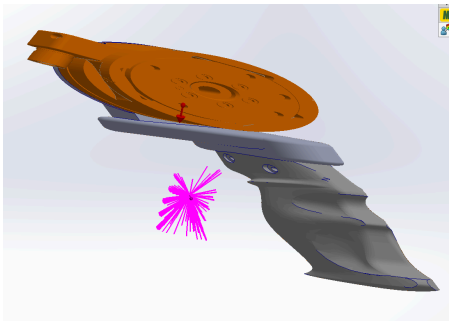
First 20 N load (X) (like a weird “twist”):



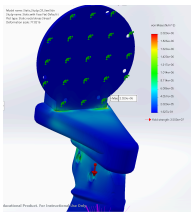
Bolt Connection Static Attempt:



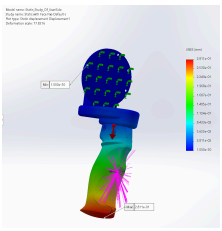
Unedited Results for first and only bolt connection attempt:



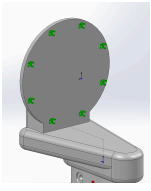
Whole face fixed, 20N in X, vonMises:



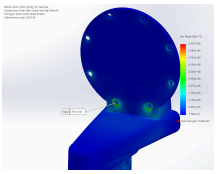
Whole face fixed, 20N in X, displacement:



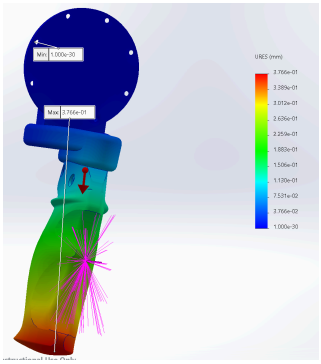
Setup for fixed screw holes:



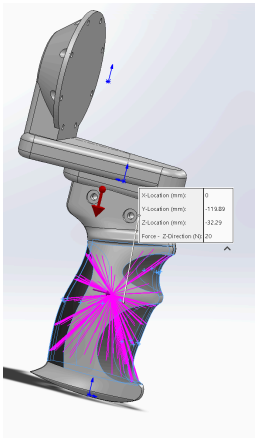
Screws fixed, 20N in X, vonMises:



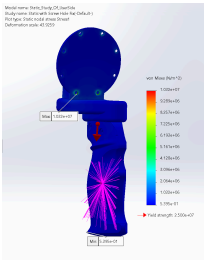
Whole face fixed, 20N in X, vonMises:



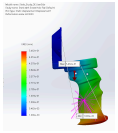
Setup for the Z axis 20[N] load (like a push into the key):



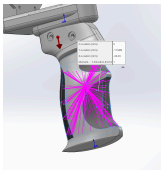
Screws fixed, 20N in Z, vonMises:



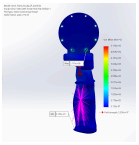
Screws fixed, 20N in Z, displacement:



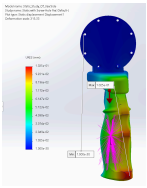
Setup for a twist of the handle (1 [Nm] in Y)



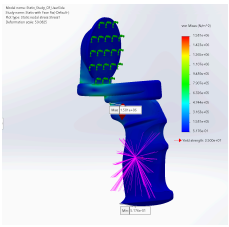
Screws fixed, 1Nm in Y, vonMises:



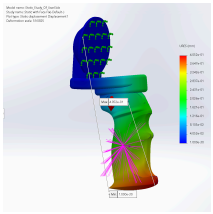
Screws fixed, 1Nm in Y, displacement:



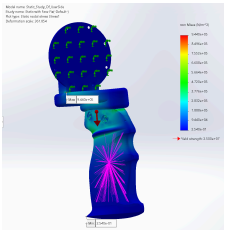
Whole face fixed, 20N in Z, vonMises:



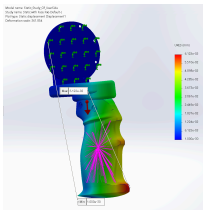
Whole face fixed, 20N in Z, displacement:



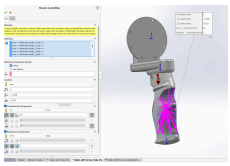
Whole face fixed, 1Nm in Y, vonMises:



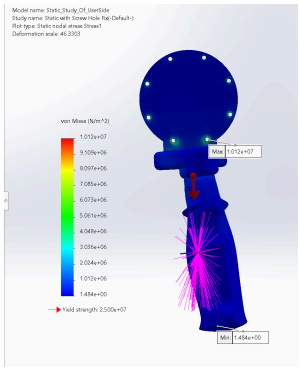
Whole face fixed, 1Nm in Y, displacement:



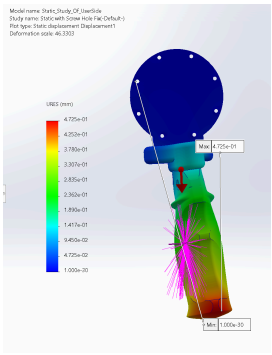
Setup for multiple loads (1 Nm in Y and 20 N in X):



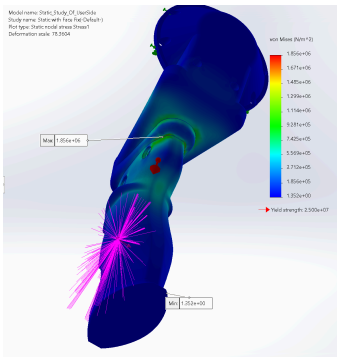
Screws fixed, 20N in X + 1Nm in Y, vonMises:



Screws fixed, 20N in X + 1Nm in Y, displacement:



Whole face fixed, 20N in X + 1Nm in Y, vonMises:



Whole face fixed, 20N in X + 1Nm in Y, displacement:

